

The empirical recurrence for OEIS A234164: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A234164 records “Empirical recurrence of order 80”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 625$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 80, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 80 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A234164 is “Number of $(n+1) \times (3+1)$ 0..4 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 2 (constant-stress 1×1 tilings)”. It has offset 1 and begins

4296, 32070, 236876, 1773484, 13227132, 99877680, 751361300, 5729616704, ...

The entry states:

Empirical recurrence of order 80 (see link above).

As of the “Last modified” line on the live entry (Jun 19 2022 01:45:02, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 625$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 625$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1250$ exact terms of the model returns order 80 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{80} c_i a(n-i),$$

c_1	27	c_{21}	−117071011132992108761	c_{41}	−12539278625071756222148636619
c_2	360	c_{22}	−6099793702730358569608	c_{42}	−17777212288976416108031895672
c_3	−15700	c_{23}	5315681944550163636644	c_{43}	20719832939370408253247142284
c_4	−17748	c_{24}	123934004190650062201114	c_{44}	591674998781201869213804906026
c_5	4139770	c_{25}	−135568930069980495988044	c_{45}	208654271724375598035504944603
c_6	−13910721	c_{26}	−2062802902620037493727699	c_{46}	−164574981494065854486388711889
c_7	−653691433	c_{27}	2447826776387540787088897	c_{47}	−28738689443661972972730302618
c_8	3990219713	c_{28}	28373323699874350307143630	c_{48}	3815947836407120403662291146210
c_9	68687068993	c_{29}	−33475706616029742979860778	c_{49}	1153657135245611510896634131653
c_{10}	−577896911764	c_{30}	−324630548665170583084821070	c_{50}	−735274123274290793187674434351
c_{11}	−5035695693632	c_{31}	356944605130267426443929018	c_{51}	−308922485830586530497838028774
c_{12}	55122510997803	c_{32}	3103588852966590187016338908	c_{52}	1172886922679491432519078702168
c_{13}	261221178736825	c_{33}	−3007285783538674162843888092	c_{53}	6224648418309200709768977673600
c_{14}	−3780993311885265	c_{34}	−24864016391285686649675743044	c_{54}	−154159355759505887784353987670
c_{15}	−9372085064695813	c_{35}	20088854325241189460823633396	c_{55}	−982893901474285763150876361330
c_{16}	194909978514577662	c_{36}	167174837883476846431874547916	c_{56}	1659341867514370937450617408197
c_{17}	206988682599345712	c_{37}	−105710836266985708827655542876	c_{57}	1237286857026702706809436547329
c_{18}	−7759644048651289041	c_{38}	−943804324690285624077985979988	c_{58}	−145060599610117654616779451611
c_{19}	−1027490137604316685	c_{39}	428337597667583298125527176844	c_{59}	−125056938169572240115600370501
c_{20}	243270030566499477891	c_{40}	4472812506776545156525721267624	c_{60}	1017708860363592988535012052240

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 80, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $80 + 4$ terms removed returns order 80 as well, so $\deg q_{\min} = 80 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $80 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 80 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A234164.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.